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## Original Research Article

## Feiji Recipe inhibits the growth of lung cancer by modulating T-cell immunity through indoleamine-2,3-dioxygenase pathway in an orthotopic implantation model

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## ABSTRACT

**Objective:** Escape from the body's immune response is a basic characteristic of lung cancer, and indoleamine-2,3-dioxygenase (IDO) plays a key role in mediating immune escape of non-small-cell lung cancer, which leads to recurrence and metastasis. Feiji Recipe, a compound Chinese herbal medicine, has the effect of stabilizing lesions and prolonging survival in patients with lung cancer. The purpose of this study was to investigate the mechanisms underlying the anticancer properties of Feiji Recipe.**Methods:** An orthotopic transplant model of mouse Lewis lung cancer, with stable expression of IDO gene, was established in C57BL/6 mice. Optical imaging was used to observe the effects of Feiji Recipe in the treatment of lung cancer *in vivo*. The effects of Feiji Recipe on the proliferation of mouse Lewis lung cancer cell line 2LL, 2LL-enhanced green fluorescent protein (2LL-EGFP) and 2LL-EGFP-IDO were investigated, and the apoptosis of T-cells was examined by 3-(4,5-dimethyl-2-thiazolyl)-2,5-diphenyl-2H-tetrazolium bromide using flow cytometry. Chemical composition of Feiji Recipe was validated by high-performance liquid chromatography.**Results:** Compared to the control group, the survival of animals treated with Feiji Recipe was significantly prolonged ( $P = 0.0074$ ), and the IDO protein level decreased ( $P = 0.0072$ ); moreover, the percentages of CD4<sup>+</sup>CD25<sup>+</sup> T-cells and Foxp3<sup>+</sup> T-cells were significantly decreased ( $P < 0.05$ ). The molecular mechanism of Feiji Recipe against lung cancer may relate to the regulation of immune cells, such as T-cells and regulatory T-cells.**Conclusion:** The molecular mechanism of Feiji Recipe in treatment of lung cancer is to restore the function of T-cells in the cancer microenvironment through interfering with the IDO pathway.Please cite this article as: Luo B, Que ZJ, Zhou ZY, Wang Q, Dong CS, Jiang Y, Hu B, Shi H, Jin Y, Liu JW, Li HG, Wang L, Tian JH. Feiji Recipe inhibits the growth of lung cancer by modulating T-cell immunity through indoleamine-2,3-dioxygenase pathway in an orthotopic implantation model. *J Integr Med*. 2018; 16(4): 283–289

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## 1. Introduction

Lung cancer is a major public health problem worldwide, with non-small-cell lung cancer (NSCLC) being the most frequent and

aggressive type [1,2]. Metastasis remains the cause of 90% of deaths from solid tumors [3], but the mechanism of lung cancer metastasis is still unknown. Immune escape and acquisition of “tolerance” by tumor cells are essential to tumor growth and progression, which are also the key factors of metastasis. Recent studies suggest that the immune-regulatory enzyme indoleamine-2,3-dioxygenase (IDO) may contribute to tumor-induced tolerance [4]. IDO is an enzyme that catalyzes the first and rate-limiting step in the kynurenine pathway of tryptophan catabolism [5]. The

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expression of IDO in tumor cells causes depletion of tryptophan in the microenvironment, as well as production of kynurenine, causing growth arrest and apoptosis of T lymphocytes, or natural killer cells, and inducing the generation of regulatory T-cells (T-regs) [6].

Although surgery, chemotherapy, radiotherapy, immunotherapy and gene therapy have made great progress, there is still a lack of effective means for the treatment of metastasis of lung cancer [7]. As a kind of “green therapy,” traditional Chinese medicine can have antitumor effects, but their mechanisms are still unknown. Feiji Recipe is a classical herbal recipe that has been used in clinical practice for more than 30 years. Feiji Recipe can stabilize the tumor burden, relieve the clinical symptoms, improve patients’ quality of life and alleviate side effects of chemotherapy [8]. Besides that, Feiji Recipe can decrease the levels of inhibitory immune cells of CD4<sup>+</sup>CD25<sup>+</sup> T-regs, the inhibitory immune factors of vascular endothelial growth factor, soluble CD44 variant-6, transforming growth factor- $\beta$ 1 (TGF- $\beta$ 1) and interleukin-10 (IL-10) in mice [9], and CD4<sup>+</sup>CD25<sup>+</sup> T-reg-secreting TGF- $\beta$ 1, IL-10 and other immune inhibitory factors which were crucial for immune suppression [10]. Feiji Recipe was effective in restoring immunosurveillance and intervening with immune escape of lung cancer [9]. But, the molecular mechanism is still unclear.

Therefore, in this work, we provided a new mouse model for studying the role of IDO in lung cancer. We directly injected an IDO-expressing lung cancer cell line into the left lung of C57BL/6 mice and established the orthotopic transplant model of murine Lewis lung cancer to explore the tumor-suppressive features of Feiji Recipe. We analyzed the percentage of various immune cell subsets, finding that Feiji Recipe restrains tumor growth *in vivo* through reduced expression of IDO. More importantly, we aimed to figure out the relationship between the tumor-suppressive effect of Feiji Recipe and IDO-mediated immune escape in lung cancer.

## 2. Materials and methods

### 2.1. Reagents and cell lines

Chemicals of 3-(4,5-dimethyl-2-thiazolyl)-2,5-diphenyl-2-H-tetrazolium bromide (MTT) and 1-methyl-tryptophanand (1-MT) were obtained from Sigma–Aldrich. Feiji Recipe was obtained from Longhua Hospital. Monoclonal anti-IDO and anti- $\beta$ -actin antibodies were from Santa Cruz Biotechnology. The mouse Lewis lung cancer cell line 2LL was purchased from the Cell Bank of the Chinese Academy of Sciences and was cultured in Dulbecco’s modified Eagle medium (DMEM; Sigma–Aldrich) supplemented with 10% fetal bovine serum (FBS; Sigma–Aldrich) and penicillin (100 U/mL)-streptomycin (100  $\mu$ g/mL; Sigma–Aldrich). Cells were grown at 37 °C in a humidified atmosphere with 5% CO<sub>2</sub>.

### 2.2. Sample preparation

Feiji Recipe consists of *Astragalus mongholicus* (Shenghuangqi), *Radix Glehniae* (Beishashen), *Liriope spicata* (Maimendong), *Asparagus cochinchinensis* (Tiandong), *Poria cocos* (Fuling), *Selaginella doederleinii* (Shishangbai), *Salvia chinensis* Benth (Shijianchuan), *Houttuynia cordata* Thunb (Vuxingcao) and *Paris polyphylla* (Qiyeyizhihua); all crude herbs were harvested according to “Good Agricultural Practice for Chinese Crude Drugs.” The processed herbs and the decoction were prepared by complying with the Good Manufacturing Practice specific to Chinese herbal medicine [8].

### 2.3. High-performance liquid chromatography analysis

IDO enzyme activity was evaluated by measuring the concentrations of tryptophan and kynurenine in the conditioned medium, using a high-performance liquid chromatography (HPLC) system. A Unitary C18 column (4.6 mm  $\times$  150 mm, 5  $\mu$ m, Acchrom) was used for reversed-phase liquid chromatography (RPLC) analysis. The column temperature was set at 25 °C. Injection volumes were 50  $\mu$ L and 1  $\mu$ L. The flow rate was 1.0 mL/min. Mobile phase A was H<sub>2</sub>O (0.1% trifluoroacetic acid (TFA)) and B was acetonitrile (0.1% TFA). Gradient was as follows: 0–5 min, A/B, (95/5, v/v)  $\rightarrow$  (90/10, v/v), 5–40 min, A/B, (90/10, v/v)  $\rightarrow$  (60/40, v/v), 40–55 min, A/B, (60/40, v/v)  $\rightarrow$  (5/95, v/v). The detecting wavelength was set at 254 nm.

### 2.4. Mice

Male C57BL/6 mice weighing (18  $\pm$  2) g, 6 weeks of age, were obtained from the Experimental Animal Center of Fudan University and bred in the Longhua Hospital, Shanghai University of Traditional Chinese Medicine. Under the pathogen-free environment, all animals were cared for and anesthetized before experimentation in accordance with the *Guide for the Care and Use of Laboratory Animals*.

### 2.5. MTT assay

In our knowledge, intact tissue implanted orthotopically which would have enabled metastasis to occur has been better detected by green fluorescent protein (GFP) than luciferase [11]. We constructed the 2LL cell line that overexpressed IDO and carried the GFP. The inhibition of the proliferation of 2LL, 2LL-enhanced green fluorescent protein (2LL-EGFP) and 2LL-EGFP-IDO cells by Feiji Recipe was assessed using the colorimetric MTT assay as described previously [12]. Cells were plated at a density of 5000 cells/well in 96-well plates and treated with Feiji Recipe in a range of concentrations from 5% to 20% for 72 h, prior to the assay for cell viability by MTT. All experiments were biologically repeated in triplicate to verify consistency of the results obtained.

### 2.6. Orthotopic xenografts

For orthotopic implantation, 2LL-EGFP-IDO cells (5  $\times$  10<sup>5</sup> cells/mL, 0.2 mL) were injected into the lung of the C57BL/6 mice under anesthesia ( $n$  = 10 per group). Tumor progression of 2LL-EGFP-IDO xenografts was detected with bioluminescent signaling during the course of study [13]. All experiments were carried out under the approval of the administrative panel on laboratory animal care of Longhua Hospital.

### 2.7. In vivo optical imaging

Before imaging, the mice were anesthetized with phenobarbital sodium. Exposure time for bioluminescence imaging was in the range of 30 s to 5 min. Fluorescence signals were simultaneously quantified using Kodak *In-Vivo* Imaging System FX Pro on days 4, 8 and 12 after orthotopic implantation.

### 2.8. Splenocyte isolation

Fresh lymphocyte suspension from male C57BL/6 mice was prepared prior to each experiment as previously described [14]. The mice were exsanguinated, and their spleens were immediately excised. Cells derived from the spleens were sequentially passed through a 100-mesh screen to harvest the mononuclear lymphocytes. The erythrocytes were lysed with Gey’s reagent for 5 min

and then washed with sterile phosphate-buffered saline. Mononuclear lymphocytes were isolated by buoyant density centrifugation (10 min at  $3000 \times g$ ) and resuspended at  $1 \times 10^6$  cells/mL in DMEM supplemented with 10% (v/v) FBS. The cells were seeded in 6 new culture plates at  $2 \times 10^5$  lymphocytes per well and incubated at  $37^\circ\text{C}$  in a humidified atmosphere of 5%  $\text{CO}_2$  for the indicated period.

### 2.9. Coculture of 2LL-EGFP-IDO cells with T-cells

As previously described,  $1 \times 10^4$  2LL-EGFP-IDO cells per well were plated in a 12-well plate [15]. After the cells attached to the bottom, they were cocultured with  $1 \times 10^5$  T-cells in DMEM containing 10% FBS and 200 U/mL human recombinant IL-2 (Shuanglu, Beijing, China) at  $37^\circ\text{C}$  and 5%  $\text{CO}_2$  for 48 h. Cultures of 2LL-EGFP cells with T-cells were performed as controls. After incubation, the nonadherent T-cells were collected and analyzed by flow cytometry.

### 2.10. Apoptosis of T-cells and the proportion of Tregs were detected by flow cytometry

The extent of apoptosis was measured through the annexin V-fluorescein isothiocyanate (FITC) apoptosis detection kit (Invitrogen Corporation, CA, USA) according to the manufacturer's instructions. After coculture, T-cells were collected and incubated with annexin V-FITC and propidium iodide. Finally, the cells were analyzed using a fluorescence-activated cell sorter (FACS; Becton Dickinson, NJ, USA).

Freshly isolated T-cells were cocultured with 2LL-EGFP-IDO cells and treated with Feiji Recipe. After 48 h, T-cells were then collected, fixed, permeabilized (eBioscience) and stained with fluorescent antibody, according to the manufacturer's instructions. The stained cells were analyzed using FACS.

### 2.11. Statistical analysis

Results are expressed as mean  $\pm$  standard deviation. Statistical significance was assessed using Student's *t* test. Statistical analysis of the luciferase signal was performed using the Mann–Whitney *U* test to evaluate the significance of differences between groups. The Kaplan–Meier method was used to generate survival curves, and the comparison was done using log-rank tests.  $P \leq 0.05$  was considered statistically significant. Cell experiments *in vitro* were repeated three times.

## 3. Results

### 3.1. Phytochemical analysis of Feiji Recipe by HPLC

In order to control the quality of Feiji Recipe, a fingerprint analysis was developed using RPLC. As shown in Fig. 1, many peaks were detected, indicating the abundance of chemical components. Among the constituents, there were many differences in polarity, as well as relative concentration. Some components of Feiji Recipe were identifiable by comparing chromatograms with known standards, including the total flavonoids of *Astragalus*, astragaloside and *Paris saponin* (data not shown).

### 3.2. Effects of Feiji Recipe on tumor cell growth

To investigate whether Feiji Recipe could inhibit the tumor progression *in vitro*, we used the MTT assay. As shown in Fig. 2, Feiji Recipe reduced 2LL, 2LL-EGFP and 2LL-EGFP-IDO cell proliferation in a concentration-dependent manner. Although the Feiji Recipe

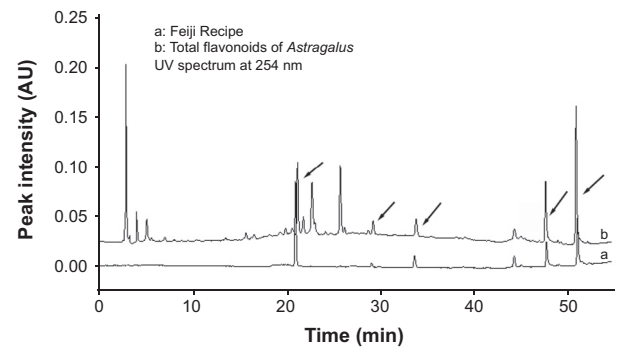


Fig. 1. High-performance liquid chromatography fingerprint analysis of Feiji Recipe. We have detected the total flavonoids of *Astragalus* in the Feiji Recipe. UV: ultraviolet.

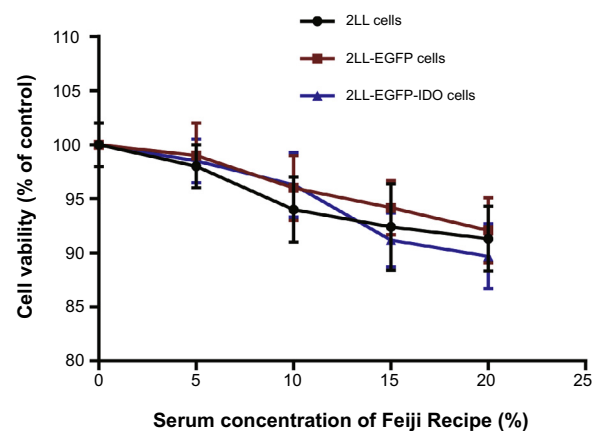


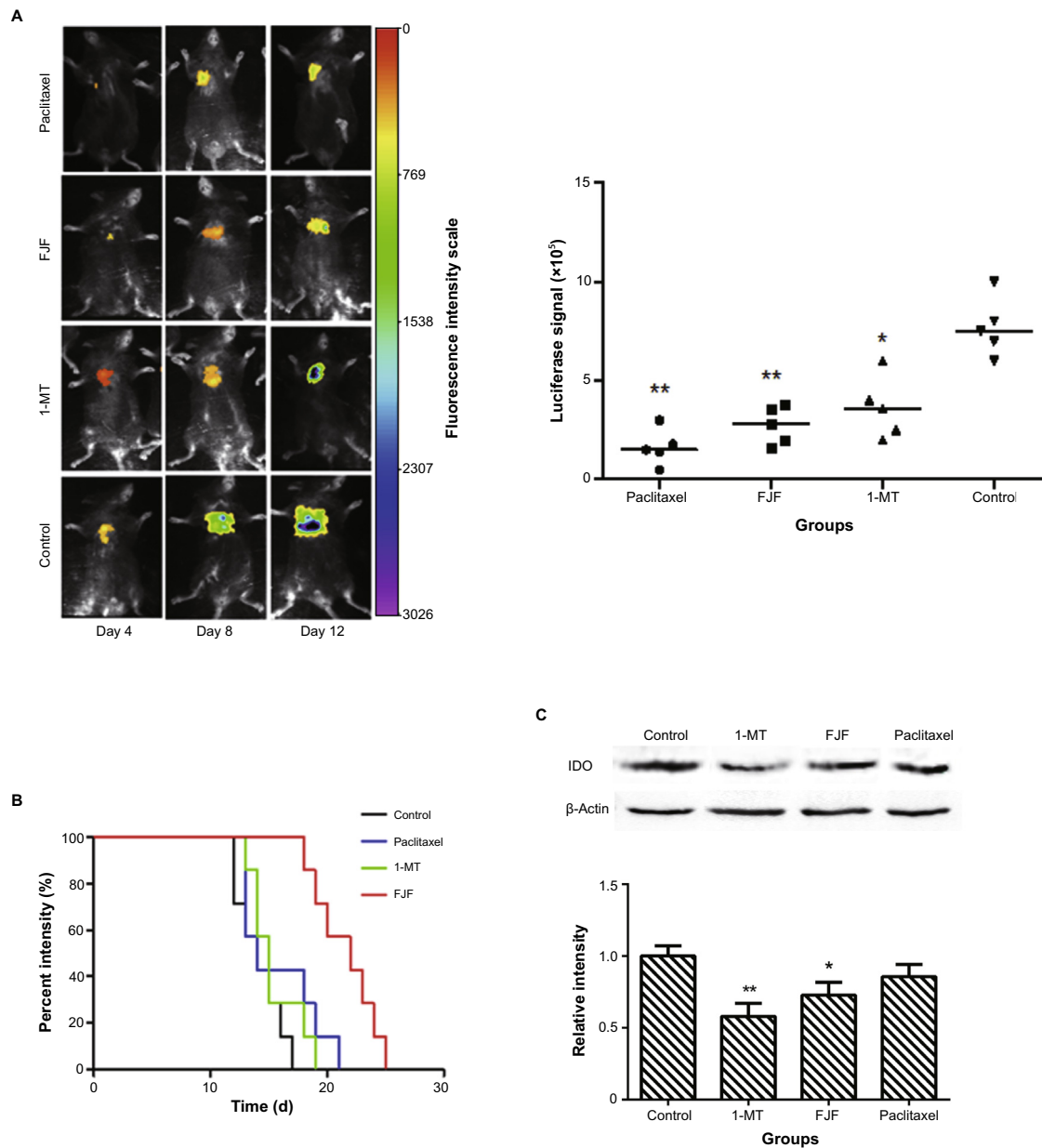
Fig. 2. Effects of Feiji Recipe on cell proliferation of 2LL, 2LL-EGFP and 2LL-EGFP-IDO cells *in vitro*. EGFP: enhanced green fluorescent protein; IDO: indoleamine-2,3-dioxygenase. Data are expressed as mean  $\pm$  standard deviation of three separate experiments.

drug serum concentration reached 20%, the inhibitory rate of cell proliferation was only 10%. Therefore, we speculate that Feiji Recipe has other effects in inhibiting tumor proliferation and stabilizing tumor focus.

### 3.3. Feiji Recipe significantly inhibited tumor growth and prolonged survival by reducing IDO expression *in vivo*

To investigate the role of Feiji Recipe in tumor progression, *in vivo* tumor growth of 2LL-EGFP-IDO cells was established in a C57BL/6 orthotopic mouse model. Not only did the fluorescence intensity of tumors in the Feiji Recipe treatment group significantly decrease at 4-, 8-, and 12-day time points (Mann–Whitney *U* test; Fig. 3A), but also the orthotopic mouse survival was significantly prolonged ( $P = 0.0074$ , log-rank test) when compared with control group (Fig. 3B). Additionally, the IDO inhibitor 1-MT and paclitaxel also significantly inhibited tumor growth and prolonged the survival of mice. Although paclitaxel has the best antitumor effect, the survival time of the mice was worse than that in the Feiji Recipe group due to toxic side effects.

Based on the MTT results, we found that Feiji Recipe did not directly inhibit tumor cell growth, which means it may have other effects, such as immunoregulation. Therefore we used Western blot to determine the effect of Feiji Recipe on the expression of IDO *in vivo*. In the Feiji Recipe treatment group, IDO protein level decreased ( $P = 0.0072$ ) compared with the control group (Fig. 3C).



**Fig. 3.** Antitumor effect of Feiji Recipe in 2LL-EGFP-IDO cell xenograft model. A: *In vivo* fluorescence imaging of 2LL-EGFP-IDO cells implanted orthotopically in C57BL/6 mice at days 4, 8 and 12 postimplantation; B: Survival curves for 2LL-EGFP-IDO-xenografted C57BL/6 mice treated with vehicle, paclitaxel, 1-MT or Feiji Recipe; C: IDO protein in tumor tissue was assayed by Western blot. EGFP: enhanced green fluorescent protein; FJF: Feiji Recipe; IDO: indoleamine-2,3-dioxygenase; 1-MT: 1-methyl-tryptophan. Statistical analysis was performed using the Mann-Whitney *U* test to evaluate the significance of differences between groups. Compared to control, \**P* < 0.05, \*\**P* < 0.01.

This result indicated that the antitumor activity of Feiji Recipe operated by regulating the expression of IDO.

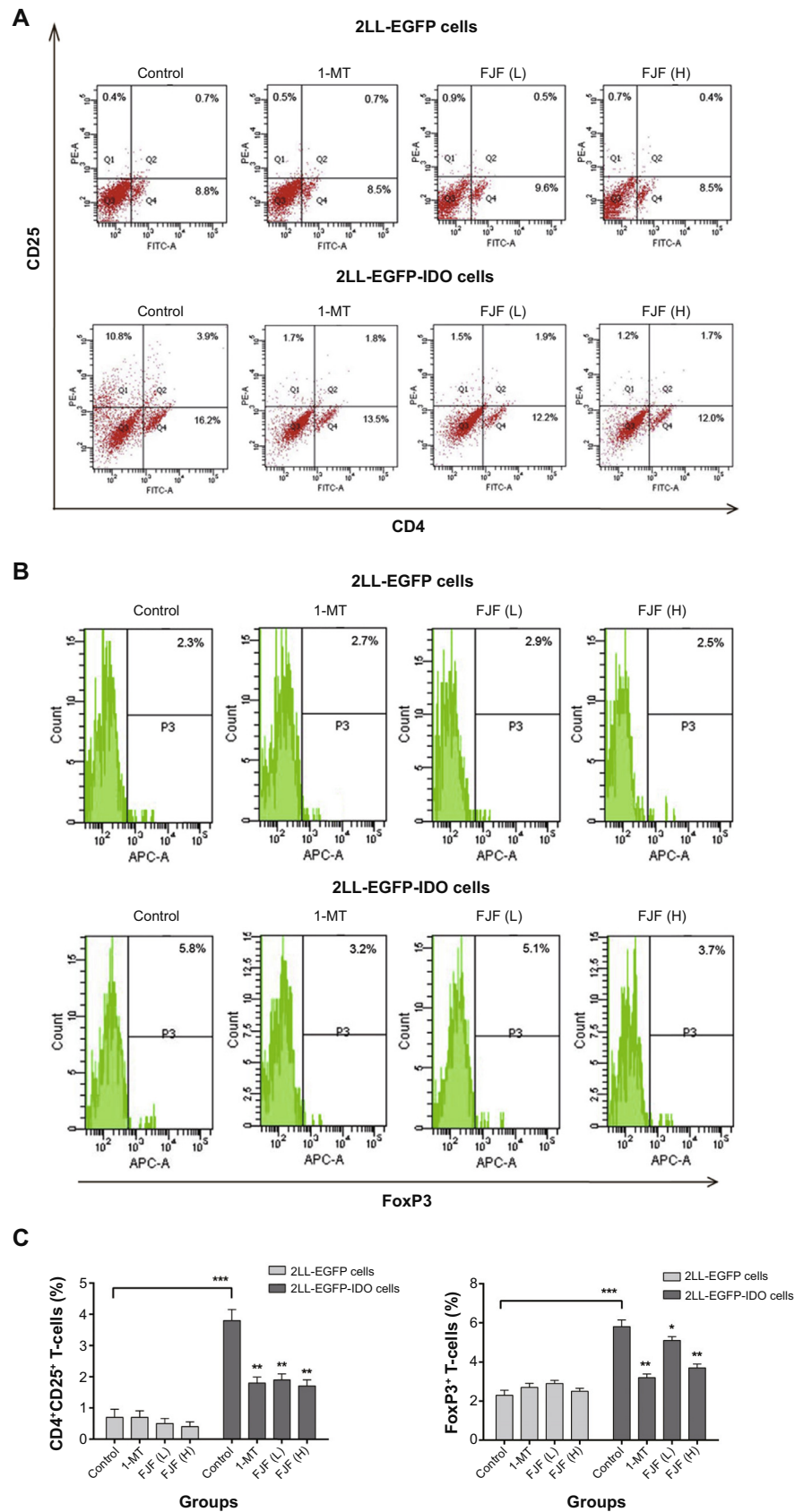
#### 3.4. Treatment with Feiji Recipe downregulated the percentage of T-reg<sub>s</sub> and inhibited apoptosis of T-cells *in vitro*

To evaluate whether Feiji Recipe regulates lymphocytes via IDO, 2LL-EGFP-IDO cells were cocultured with splenocytes, and then treated with Feiji Recipe. As shown in Fig. 4, IDO group had higher percentages of CD4<sup>+</sup>CD25<sup>+</sup> T-cells and Foxp3<sup>+</sup> T-cells compared to control group. After treatment with Feiji Recipe, the percentages of CD4<sup>+</sup>CD25<sup>+</sup> T-cells and Foxp3<sup>+</sup> T-cells significantly decreased, but no effect was observed in the 2LL-EGFP cell group.

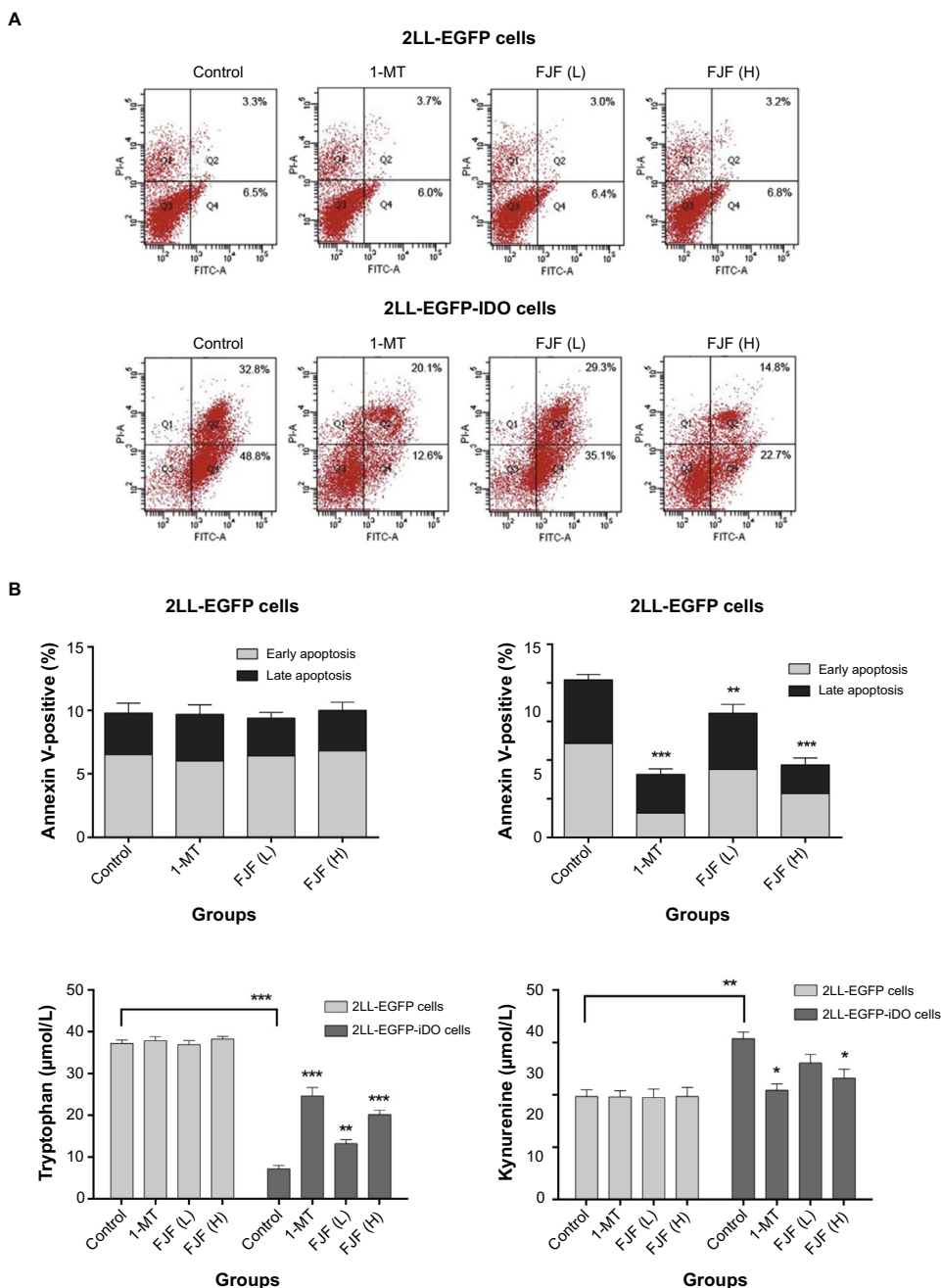
#### 3.5. Feiji Recipe significantly inhibited the production of kynurenine

IDO has also been known to stimulate T-cell apoptosis [16]. Therefore, we performed apoptosis assays to examine the effects of Feiji Recipe on apoptosis of T-cells using annexin V-FITC staining, followed by flow cytometry analysis. As shown in Fig. 5A, coculture of 2LL-EGFP-IDO cells with T-cells significantly increased T-cell apoptosis, compared to the control group. When treated with Feiji Recipe, the apoptosis of T-cells significantly decreased in the IDO group. This result showed that Feiji Recipe reduced the apoptosis of T-cells, possibly through IDO.

In order to study the effects of Feiji Recipe on the IDO enzyme activity, we assayed the levels of tryptophan and kynurenine in coculture medium. HPLC data showed that coculture medium from



**Fig. 4.** Feiji Recipe regulates T-cells and T-regs by inhibiting IDO. A: Coculture of T-cells with 2LL-EGFP-IDO cells increased the percentage of CD4<sup>+</sup>CD25<sup>+</sup> T-cells more than that with 2LL-EGFP cells and the addition of 1-MT or Feiji Recipe significantly reduced the percentage of CD4<sup>+</sup>CD25<sup>+</sup> T-cells; B: The percentage of FoxP3<sup>+</sup> T-cells, gated on CD4<sup>+</sup> T-cells, has the same results; C: There are two bar charts showing the above phenomenon. APC: antigen-presenting cell; EGFP: enhanced green fluorescent protein; FJF (H/L): Feiji Recipe of high/low concentration; IDO: indoleamine-2,3-dioxygenase; 1-MT: 1-methyl-tryptophan. Compared to control, \**P* < 0.05, \*\**P* < 0.01, \*\*\**P* < 0.001.



**Fig. 5.** The effects of Feiji Recipe on T-cell apoptosis and release of kynurenine. A: Feiji Recipe significantly inhibited apoptosis in T-cells. B: Kynurenine concentration released by 2LL-EGFP-IDO cells coculture system after treatment with Feiji Recipe and 1-MT measured after 48 h by high-performance liquid chromatography. EGFP: enhanced green fluorescent protein; FITC: fluorescein isothiocyanate; FJF (H/L): Feiji Recipe of high/low concentration; IDO: indoleamine-2,3-dioxygenase; 1-MT: 1-methyl-tryptophan. Data are presented as mean  $\pm$  standard error of the mean ( $n = 3$ ). Compared to control, \* $P < 0.05$ , \*\* $P < 0.01$ , \*\*\* $P < 0.001$ .

2LL-EGFP-IDO cells treated with Feiji Recipe exhibited a lower level of kynurenine and higher level of tryptophan compared to the control group (Fig. 5B). This result suggests that Feiji Recipe reduced the apoptosis of T-cells by inhibiting IDO production of kynurenine, through downregulation of the expression of IDO in the coculture system.

#### 4. Discussion

NSCLC is a malignant disease with a poor prognosis when locally advanced or metastasized. Despite advances in surgery, chemotherapy, radiotherapy and targeted therapy, many patients

do not benefit from these therapies. One potential new treatment is cancer immunotherapy. Inhibition of programmed cell death protein 1 (PD1)/programmed cell death ligand 1 (PDL1) is an interesting therapeutic target which restores the immune system against tumor cells. Anti-PD1 and anti-PDL1 antibodies have been developed to restore the immune system in solid cancers, including NSCLC; patients who will derive the most benefit from this therapy have high expression of PDL1 [17]. Therefore, it is important to reveal the key signaling pathways of the tumor microenvironment immune suppression and intervention, enabling clinicians to achieve accurate treatment and improve the curative effect of the therapy.

The past decade has witnessed an astonishing outpouring of research on IDO. IDO has recently been recognized as an important factor in immune regulation and development of immune tolerance in the tumor microenvironment. Therefore, the regulation of its expression may provide a strategy for enhancing immune editing [18]. A recent report suggested that higher expression of IDO in tumor tissue and lower number of CD8<sup>+</sup> tumor-infiltrating lymphocytes contributed to a lower rate of survival in NSCLC patients [17]. Reducing tumor cell expression of the immunosuppressive enzyme IDO could activate CD8<sup>+</sup> T-cells and induce regulatory T-cell apoptosis [19,20].

Feiji Recipe is composed of 9 traditional Chinese herbal medicines, and has been proven to be effective in treating NSCLC and inhibiting tumor cell metastasis in the clinical setting [8]. Prior studies showed that Feiji Recipe can inhibit the proliferation of Lewis, SPC-A-1 and LAX-8 cells [21]. Here we showed Feiji Recipe drug serum did not significantly inhibit the proliferation of 2LL, 2LL-EGFP and 2LL-EGFP-IDO cells; however in the *in situ* transplantation tumor model of lung cancer, Feiji Recipe inhibited tumor progression and prolonged survival, suggesting that Feiji Recipe may play an antitumor effect by regulating the immune response. The IDO pathway generates tryptophan degradation, kynurenines and accumulation of other downstream catabolites. This results in immunosuppression by enhancing the apoptosis of T-cells and increasing the proportion of T-regs [22,23]. It is also well accepted that IDO inhibits the antitumor immunity of cytotoxic T-cells by promoting cell cycle arrest and apoptosis [22]. Many studies have demonstrated that small-molecule inhibitors of IDO could disrupt the immunosuppressive enzyme, and then stimulate tumor immunity [24], inhibit tumor growth [25], reverse multidrug resistance and reduce the occurrence of tumor via immune escape [5,26]. Compared with *in vivo* studies, we found that Feiji Recipe not only inhibited the apoptosis of T-cells, but also decreased the proportion of T-regs by downregulating the IDO level in an *in vitro* coculture system. HPLC data confirmed that the concentration of kynurenine in the coculture medium was downregulated by Feiji Recipe, and tryptophan correspondingly increased compared to the control group. These results are consistent with previous research stating that the antitumor properties of Feiji Recipe may be based, to a significant degree, on the disruption of IDO expression.

## 5. Conclusions

We confirmed the capability of Feiji Recipe to inhibit tumor progression and prolong survival in a mouse orthotopic lung tumor model. We also demonstrated that Feiji Recipe effectively inhibited apoptosis of T-cells and decreased the proportion of T-regs. IDO helped to create a tolerogenic milieu by directly inducing T-cell apoptosis and enhancing T-regs-mediated immunosuppression. We demonstrated that IDO expression was downregulated by Feiji Recipe. This study provides an experimental basis for further study into the mechanism underlying the interaction between Feiji Recipe and IDO-mediated tumor immune escape.

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## Conflict of interests

The authors have declared no competing interests exist.

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